BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 2 Practical No: 2

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Subject Name : Deep Learning

Problem Statement

Design and implement a Multilayer Perceptron (MLP) for classification of the Wine dataset and evaluate its performance using accuracy and a confusion matrix.

Algorithm

1. Start the program.
2. Import the required libraries: NumPy, Matplotlib, Seaborn, TensorFlow, Keras, and Scikit-learn.
3. Load the Wine dataset using load_wine().
4. Separate the dataset into input features (X) and target labels (y).
5. Display the shape of the dataset, feature names, and target class names.
6. Split the dataset into training and testing sets using train_test_split().
7. Standardize the training and testing datasets using StandardScaler.
8. Create a Multilayer Perceptron (MLP) model with an input layer, two hidden layers using ReLU activation, and an output layer using Softmax activation.
9. Compile the model using the Adam optimizer and Sparse Categorical Crossentropy loss function.
10. Train the model using the training dataset for 50 epochs.
11. Plot the training and validation accuracy graph.
12. Plot the training and validation loss graph.
13. Evaluate the trained model using the testing dataset.
14. Predict the class labels for the testing dataset.

15. Calculate the classification accuracy.
16. Generate and display the confusion matrix.
17. Display the classification report.
18. Stop the program.

Code :

```
history = model.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=8,
    validation_split=0.2
)

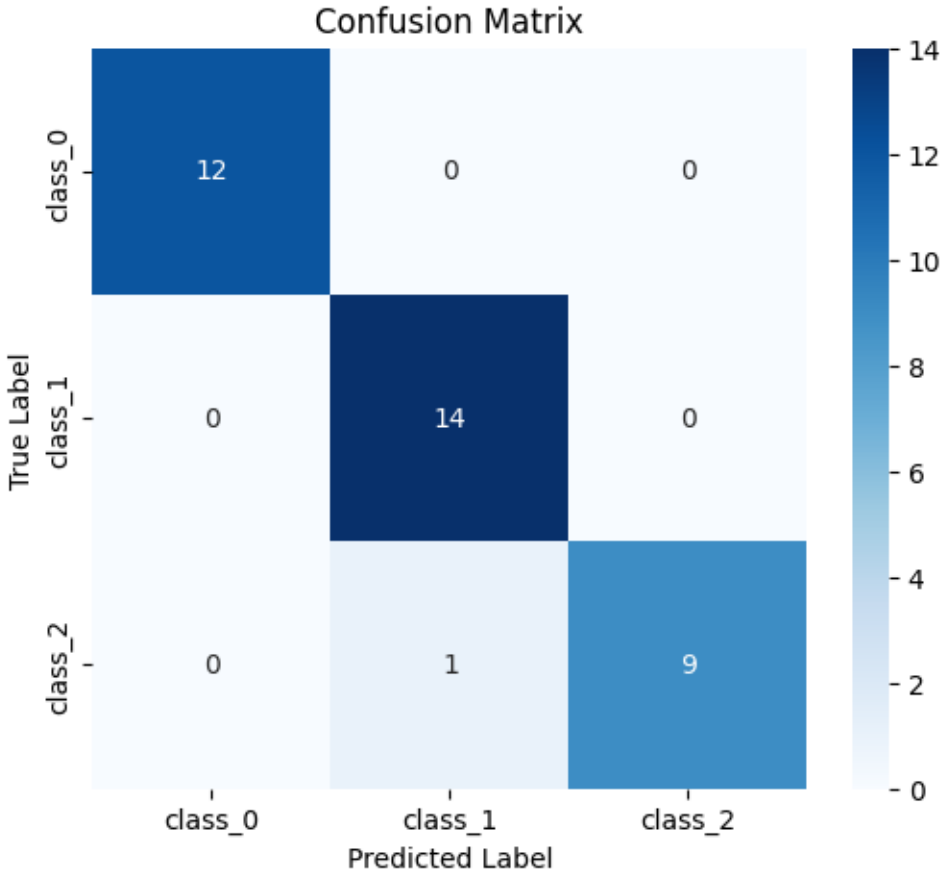
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Accuracy:", accuracy)

y_pred = model.predict(X_test)
y_pred = np.argmax(y_pred, axis=1)

cm = confusion_matrix(y_test, y_pred)
```

Output

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Final Accuracy: 97.22%